European Journal of Trauma & Dissociation 9 (2025) 100563

Contents lists available at ScienceDirect

European Journal of Trauma & Dissociation

journal homepage: www.elsevier.com/locate/ejtd

Review Article

Posttraumatic growth in children and adolescents induced by the COVID-19 pandemic: A scoping review

Alexandra Tamiolaki [a,*], Argyroula Kalaitzaki [b]

[a] Ph.D., Department of Social Work, Hellenic Mediterranean University, Laboratory of Interdisciplinary Approaches for the Enhancement of Quality of Life, School of Health Sciences, Heraklion, Crete, Greece

[b] Associate Professor of Clinical Psychology, Social Work Department, Head of Laboratory of Interdisciplinary Approaches for the Enhancement of Quality of Life, School of Health Sciences, Affiliated Researcher of the University Research Centre 'Institute of AgriFood and Life Sciences', Hellenic Mediterranean University, Heraklion, Crete, Greece

ARTICLE INFO

Keywords:
Children
Adolescence
Pandemic
Mental health
Posttraumatic growth

ABSTRACT

The COVID-19 pandemic, a highly challenging life experience, may also be a chance for posttraumatic growth (PTG). This scoping review aimed to summarise the available evidence on PTG among children and adolescents induced by COVID-19 by reporting rates and identifying potential influencing factors. The literature search yielded twenty-one articles from eleven countries which were eventually included. Findings revealed heterogenous PTG prevalence rates, ranging from low to moderate/high, and scattered facilitators, including sociodemographic, psychological, and COVID-19-related factors, with relational factors (i.e., social support) consistently highlighted. Gaps in the literature were also identified, with the lack of robust longitudinal studies and the exploration of the illusory PTG being fundamental. These findings can inform public health authorities, policymakers, and clinicians to design and implement interventions to promote children's and adolescents' growth in times of similar crises.

1. Introduction

Children and adolescents may be exposed to adverse experiences such as community violence, physical maltreatment, physical and emotional abuse, neglect, familial violence, forced displacement, terrorist attacks, fire attacks, natural disasters and pandemics, all of which can have lifelong health consequences (Abbas et al., 2024; Fung et al., 2024; Madrigal Gonzaleza et al., 2024). Children and adolescents are a vulnerable age group because their cognitive and mental abilities are still developing, and they may have few inner resources and past experiences to cope with any adversity (Amjadi, 2021; Mohler-kuo et al., 2021). Already being vulnerable, children and adolescents' mental health may be further compromised during pandemics both in the short and long term (Ramadhan et al., 2020) with unknown consequences (Loades et al., 2020) and effective trauma-informed interventions may are required (Domen, 2024; Silvestre & Tarquinio, 2022; van der Hart et al., 2017).

The COVID-19 pandemic is widely regarded as one of the most significant threats to children and adolescents' mental health, with numerous studies highlighting its profound impact on their psychological well-being (e.g., Araújo et al., 2021; Xie et al., 2020). Studies in children and adolescents have shown that the COVID-19 pandemic increased the risk of various psychological symptoms and disorders, such as anxiety, depression, stress, post-traumatic stress disorder (Araújo et al., 2021; Xie et al., 2020), grief, adjustment disorder, and acute stress disorder (ASD) (Ramadhan et al., 2020; Lee et al., 2021). Family conflicts, loneliness, social isolation, physical distance, living in urban areas or communities with high rates of COVID-19 cases are some of the factors that predict worse mental health outcomes (Octavius et al., 2020; Yue et al., 2020). Particularly vulnerable appear to be girls, pupils diagnosed with COVID-19, those with mental health difficulties, 'special educational needs and disabilities' and/or neurodevelopmental disorders (Lee et al., 2021; Ma et al., 2021; Panchal et al., 2021).

Even though the overwhelming evidence suggests that catastrophic events such as the COVID-19 pandemic undoubtedly had adverse effects on mental health, some studies have timidly shown that positive psychological outcomes are also possible (Muratori & Ciacchini, 2020; Zhou et al., 2021). One's ability to quickly recover from trauma or highly stressful and challenging life situations and emerge stronger has mostly

* Corresponding author.
E-mail addresses: alexandratmk93@gmail.com (A. Tamiolaki), akalaitzaki@hmu.gr (A. Kalaitzaki).

https://doi.org/10.1016/j.ejtd.2025.100563
Received 5 February 2025; Received in revised form 31 May 2025; Accepted 8 June 2025
Available online 8 June 2025
2468-7499/© 2025 Elsevier Masson SAS. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

been studied through the concept of posttraumatic growth (PTG). Posttraumatic growth (PTG) refers to the positive psychological changes that follow one's highly emotional struggle with traumatic or intensely stressful experiences (Tedeschi & Calhoun, 2004). PTG is assumed to occur in five domains, including developing more intimate relationships with others, gaining greater personal strength, discovering new possibilities and life paths, appreciating life more, and experiencing greater spiritual or existential growth (Tedeschi & Calhoun, 2004).

PTG shares conceptual similarities with related constructs such as resilience and adaptive coping, as all pertain to positive responses to adversity and psychological adjustment (Compas et al., 2001; Masten & Motti-Stefanidi, 2020; Tedeschi & Calhoun, 2004). While resilience and coping may contribute to the development of PTG, they are not equivalent to it. Resilience is generally defined as the capacity to maintain or return to baseline functioning following adversity, adaptive coping refers to the situational deployment of behavioral and cognitive strategies aimed at managing immediate stressors, and PTG denotes self-perceived positive psychological change that arises through the cognitive and emotional processing of traumatic experiences (Compas et al., 2001; Masten & Motti-Stefanidi, 2020; Tedeschi & Calhoun, 2004). In other words, PTG necessitates a qualitative shift in perspective or functioning, rather than simply enduring or adapting to stress (Tedeschi & Calhoun, 2004).

Researchers have suggested that PTG can be real or illusory (Zoellner & Maercker, 2006). Real PTG leads to long-lasting positive change through adaptive coping strategies (Tedeschi & Calhoun, 2004), whereas, illusory PTG leads to long-term distress because the use of maladaptive coping strategies may only be helpful in the short term (Kalaitzaki et al., 2022; Orille et al., 2019; Zoellner & Maercker, 2006). According to Tedeschi et al. (2018), there is no specific timeframe for PTG since individuals might experience PTG very early during the challenging event, or years after it has ended (Tedeschi et al., 2018). Studies (Frazier et al., 2001; Linley & Joseph, 2004) suggest that PTG appears to unfold between 2 weeks and 2 months, with reported levels remaining stable for 1 to 3 years. Thus, some studies measured PTG as early as 2 weeks after the beginning of a challenging event (Kleim & Ehlers, 2009), whereas others measured it at later timepoints (Grubaugh & Resick, 2007; Kleim & Ehlers, 2009).

Although time does not seem to be essential for the development of PTG, other factors seem to play an important role, such as personal (e.g., coping strategies), sociodemographic (e.g., age and gender), psychological (e.g., mental health), and relational characteristics (e.g., parental psychological characteristics) (Bernstein & Pfefferbaum, 2018; Masten & Motti-Stefanidi, 2020). According to Masten and Motti-Stefanidi (2020), adaptation to adversity is not just an individual process but one that unfolds through interactions across systems (family, community, education, and healthcare). This systems perspective aligns with developmental trauma theory, recognizing that challenging events disrupt multiple layers of a child's or adult's life, and that healing and growth require coordinated, multisystem support such as secure caregiving relationships, educational or workplace-based interventions, peer networks, and access to community mental health services (Bernstein & Pfefferbaum, 2018; Masten & Motti-Stefanidi, 2020).

Focusing on age-related criteria, although PTG is observed in both children-adolescents and adults, there seems to be a difference in its manifestation across these groups due to developmental factors. Researchers (Masten & Motti-Stefanidi, 2020; Tedeschi & Calhoun, 2004) underlined that for children and adolescents, the PTG process may be more challenging than for adults, as they are still developing the cognitive and emotional capacities required to find meaning through adversity. This developmental phase limits their ability to interpret their experiences in a positive and constructive manner, hindering their capacity to adapt in ways that promote growth.

The literature on the current state of knowledge of PTG in adults following the COVID-19 pandemic is particularly rich with reviews on community residents (e.g., Bovero et al., 2023; Henson et al., 2021) and healthcare workers (e.g., Li et al., 2024; O'Donovan & Burke, 2022), finding evidence of PTG and recognizing the associated factors (e.g., coping strategies, personality traits, social support, and medium levels of PTSD). Less is known about PTG in children and adolescents. Research on how children and adolescents fare in the aftermath of a challenging event offers compelling evidence on what makes a difference for the well-being of children (Masten & Motti-Stefanidi, 2020). Reviews published during the pandemic have examined children and adolescents' PTG and/or associated factors following different types of traumatic events, such as medical trauma, war- and terror-related trauma, and environmental trauma (Ferris & O'Brien, 2022), natural disasters (Fayaz, 2023), and cancer (Koutná & Blatný, 2020). Another review examined 22 studies on adolescents' PTG (Harmon & Venta, 2021), its moderators, and existing measures, after traumatic events such as cancer, sexual abuse, terrorist attack, parental divorce, death, and illness injury.

To our knowledge, there is no review focusing on PTG in children and adolescents as a result of their exposure to the COVID-19 pandemic, potentially because until recently, there was not sufficient published research to warrant a review. The predominance of research attention on adult populations in the field of PTG during COVID-19 has resulted in a marked scarcity of comprehensive reviews examining the developmental trajectory of PTG in children and adolescents, as well as the key predictors and facilitators of its emergence, thus revealing a critical gap in the extant literature. Therefore, the aim of this scoping review is to critically synthesize the literature exploring the prevalence rate and the associated factors of the pandemic-induced PTG in children and adolescents. The research question guiding this review is: What is the prevalence of posttraumatic growth among children and adolescents during the COVID-19 pandemic, and what factors are associated with its emergence? Studying this question is crucial to better understand posttraumatic growth (PTG) in children and adolescents. It can help develop effective interventions, guide resource allocation, and shape policies that protect young people's mental health in countries facing crises like pandemics.

2. Methodology

The scoping review framework developed by Arksey and O'Malley (2005) was followed including five stages: (1) identifying the research questions; (2) identifying relevant studies; (3) study selection; (4) charting the data; and (5) collating, summarizing and reporting the results. To identify published articles on COVID-19 pandemic-induced PTG in children and adolescents, we searched Scopus, PubMed, Google Scholar, and PsycINFO from February 2000 to February 2025. Only articles published in English were included. The search strategy used the following Boolean operators ("children" OR "adolescents") AND ("post--traumatic growth" OR "posttraumatic growth") AND ("COVID-19″ OR "pandemic") AND ("mental health" OR "psychological impact" OR "psychological outcomes"). Articles written in languages other than English, review articles, non-peer-reviewed articles, qualitative or narrative studies, commentaries, letters to the editors, abstracts, posters, case studies, or erratum, and duplicate studies published in multiple places were excluded from this review. Studies only on adult samples and not focusing on children and/or adolescents were also excluded. Although qualitative studies may have provided richer insights into subjective experiences of PTG, they were excluded as they do not provide prevalence rates

The authors performed an initial screening of the articles identified by the search strategy by reading the titles and abstracts. Reading the full text of the articles was the second step in selecting the eligible articles. Decision on the inclusion of the articles was made by both authors and any disagreements were resolved through detailed discussions. Then, the authors prepared a table of the selected articles (Table 1), which summarizes the key characteristics of included articles, including the reference (authors, year of publication), the aim of the study, the

Table 1
Summary of the Key Characteristics of Included Studies.

Authors	Aims	Study Design	PTG Instrument	Recruitment period	Sample	Country	Main results related to PTG
Bhushan et al. (2022)	This study examined the direct (i.e., death or hospitalization in the family) and the indirect exposure of COVID-19-related experiences (i.e., media reports) on children and adolescents and its subsequent relationship with PTS and PTG.	Cross-sectional survey	The Revised Post-Traumatic Growth Inventory for Children	July 2021 (after the second lockdown).	Children and adolescents (9–20 years old)	India	The results showed that 39.8 % of the participants who reported PTS also experienced PTG. Participants who had at least one or more siblings experienced higher PTG as compared to those without siblings. Those who read only electronic magazines reported significantly higher PTG than those not reading magazines.
Casali et al. (2023)	This study aimed at assessing the direct and indirect longitudinal relations between grit components (perseverance of effort and consistency of interests), life satisfaction at two time-points and PTG in school-aged students.	Longitudinal study, that examined PTG only on Time point 2.	The Revised Post-Traumatic Growth Inventory for Children	Between the last two weeks of May and the first two weeks of June 2021 (T2) (after the second lockdown).	Adolescents (11–18 years old)	Italy	The results showed that the mean level of PTG was 23.16 and perseverance was positively associated with PTG, indirectly favoring life satisfaction. Furthermore, male students reported greater PTG.
Hamama et al. (2024)	This study aimed at assessing the links between adolescents' PTG, parents' PTG, adolescents' differentiation of self, and adolescents' gender.	Cross-sectional study	The Posttraumatic Growth Inventory	Two years after the COVID-19 outbreak.	Adolescents (11–17 years old)	Israel	The results revealed a positive connection between adolescents' PTG and parents' PTG, and negative connection between adolescents' differentiation of self and adolescents' PTG/ parents' PTG. A higher level of adolescents' differentiation of self was associated with lower PTG in parents, which, in turn, was linked to lower PTG in adolescents, but this pattern was only observed in male adolescents.
Jian et al. (2022)	This study examined the PTG in adolescents and the moderating effect of self-efficacy on post-traumatic stress disorder (PTSD) symptoms and PTG.	Cross-sectional survey	The Posttraumatic Growth Inventory	From February 24 to February 28, 2020 (during the first lockdown).	Adolescents (12–18 years old)	China	The results showed that 20.6 % of the sample had PTG and a reverse U-shaped curve relationship between PTSD and PTG. Self-efficacy positively correlated with PTG and moderated the relationship between PTSD and PTG. For those with low self-efficacy, the higher the PTSD, the higher the PTG, and for those with high self-efficacy, the higher the PTSD, the lower the PTG.
Kazemi et al., 2022	This study investigated the positive changes that occurred in young adolescents' lives after the COVID-19 pandemic, and how they relate to perceived social support.	Cross-sectional study	Persian abbreviated version of the PTGI	From December 2020 to January 2021 (during the first and second lockdown).	Adolescents (12–18 years old)	Iran	The results showed moderate levels of PTG. Males had higher PTG than females. PTG scores were higher in young homeschooled adolescents whose parents worked remotely and had low income in comparison with adolescents whose parents did not work remotely. PTG scores

(*continued on next page*)

Table 1 (*continued*)

Authors	Aims	Study Design	PTG Instrument	Recruitment period	Sample	Country	Main results related to PTG
							were higher for young adolescents who did not have COVID-19 in comparison to others who experienced COVID-19.
Kim et al. (2022)	The study aimed to determine the feasibility and benefits of a videoconferencing-based home exercise programme to improve health-related quality of life (HRQOL), PTG and physical strength levels of paediatric cancer survivors during COVID-19.	Pilot Pretest–posttest experimental design	The Posttraumatic Growth Inventory	N/A	Children (9–12 years old)	Korea	The results showed that the exercise program did not significantly promote PTG in pediatric cancer survivors. At baseline, the levels of PTG were moderate to high (50.38) and after the intervention, the levels slightly decreased (49.63).
Kohlboeck et al. (2023)	The study explored whether trauma symptoms, psychopathology, COVID-19-related worries, financial and job problems and children's PTG as reported by parents can predict a higher risk of COVID-19-associated problems among children.	Cross-sectional study	PTG was measured using the open-ended question "What positive impact do you think the coronavirus crisis has/had on your child?"	Between March 2020 and July 2022 (during the first lockdown and after the second lockdown)	Children (7–13 years old)	North Tyrol (Austria) and South Tyrol (Italy)	The results showed that more PTG occurred in the "No-Risk" Cluster, who reported lower levels of trauma compared to "High-Risk" Cluster.
Lanfredi et al. (2023)	This study examined the mental health status among a group of Italian adolescents and young adults during the COVID-19 pandemic and the risk and protective factors associated with mental health.	Cross-sectional survey	The Post-traumatic Growth Inventory–Short Form	November 2021 (during the second lockdown)	Adolescents and young adults (14–25 years old)	Italy	The results showed low levels of PTG (20.95). Females reported slightly higher levels of PTG than males.
Senejko et al. (2022)	This study aimed to determine whether pandemic threats could activate development responses among adolescents and what is the role of the level of COVID-19-related stress in the process of PTG.	Cross-sectional survey	The Posttraumatic Growth Inventory	April 2020 (during the first lockdown).	Adolescents (14–20 years old)	Poland	The results showed that the pandemic's threats (to life, family, and lifestyle) were positive predictors of PTG. Personality traits (extroversion), level of stress, and gender had positive mediating effects on PTG. Females experienced greater PTG than males. Three of the four dimensions of PTG (relating to others, appreciation of life, and spiritual and existential change) were activated under pandemic conditions, but not self-perception.
Taddei et al. (2024)	The study aimed to examine (a)the univariate change in PTG, (b) the influence of internal resilience at baseline on growth, and (c) the change of growth due to the time-varying effect of COVID-19-related stress, and family- and school-satisfaction .	Longitudinal study	The Revised Post-Traumatic Growth Inventory for Children	The study was conducted during three-time points, the first in January/ February 2020 (T0), before the first COVID-19 outbreak in Italy, the second in May/June 2020 (T1) after the COVID-19 lockdown, the third in May/June 2021 (T2), and the final one in May/June 2022 (T3).	Adolescents (10–14 years old)	Italy	The results found a linear decrease in PTG over time. This decline was influenced by dynamic factors, including COVID-19-related stress and satisfaction with family and school environments. The initially elevated levels of PTG did not reflect genuine, long-term growth.
Tang et al. (2022)	This study examined whether baseline perceived emotional intelligence (EI) predicted PTG through self-	Longitudinal study	The Posttraumatic Growth Inventory	The baseline survey (T1) was conducted between February 24 and	Adolescents (12–18 years old)	China	The results of the study showed that there was a slight increase in PTG between the baseline

(*continued on next page*)

Table 1 (*continued*)

Authors	Aims	Study Design	PTG Instrument	Recruitment period	Sample	Country	Main results related to PTG
	esteem and emotional regulation (ER) in a five-month follow-up study conducted during the COVID-19 pandemic.			February 28, 2020, and the follow-up survey (T2) was conducted between July 11 and July 23, 2020 (during and after the first lockdown).			(22.4 %) and follow-up (25.0 %). Emotional intelligence predicted PTG and self-esteem. Emotional regulation played a mediating role in the relationship between self-esteem and PTG. Males were more likely to have higher PTG than females.
Tamiolaki et al. (2024)	This study aimed to promote PTG through a school-based intervention of eight sessions of 45 min each with mindfulness and character strengths practices and explore PTG predictors.	Pretest–posttest and follow up experimental design	The Revised Post-Traumatic Growth Inventory for Children	The school year 2022–2023 (after the second COVID-19 lockdown).	Children (8–10 years old)	Greece	The results showed that the children in the intervention group experienced an improvement in PTG compared to the children in the control group, and these positive changes were sustained at the follow-up. In the intervention group, the levels of PTG were low at baseline (11.33), and moderate to high after the intervention (22.70) and at the follow-up (22.47). The predictive role of mindfulness and strengths use were shown on PTG, well-being, and PTS. The intervention increased PTG in the participants of both genders and of both grades (third and fourth).
Ulset & von Soest, 2022	This study examined the prevalence of PTG during the first lockdown, the association of PTG with the pandemic experiences and with other potential risk and protective factors, and whether these associations were moderated by mental health resources and social support.	Cross-sectional survey	The Post-traumatic Growth Inventory–Short Form	From April 23 to May 8, 2020 (during the first lockdown)	Adolescents (11–16 years old)	Norway	Results showed that 9.6 % of the adolescents experienced moderate-to-high levels of PTG. Satisfaction with life, parental care, worries about the pandemic, immigrant status, close friendships, and someone to talk to were related to higher PTG. Having a parent who was working out of home was associated with less PTG. Good mental health was also associated with higher PTG for those considered to be at higher risk of adverse outcomes.
Waters et al., 2020	This study examined stress-related growth (SRG) in a sample of students returning to campus after a period of COVID-19 remote learning. Positive reappraisal, emotional processing, and strengths use were examined as mediators.	Cross-sectional survey	Abbreviated Stress-Related Growth Scale	After the first lockdown	Adolescents (11–18 years old)	Australia	The results showed that a set of cognitive, emotional, and behavioral coping skills used during remote learning predicted SRG upon school re-entry. All three coping skills significantly predicted the degree to which students reported growth. Positive reappraisal had the strongest effect, followed by strengths use and emotional processing.
Wenter et al. (2022)	This study examined the impact of the COVID-19 pandemic on children's	Concurrent Design	PTG was measured using the open-ended question "What	March 2020 (t1), December 2020 (t2), June 2021 (t3),	Children (3–13 years old)	North Tyrol (Austria)	The results of the study showed that perceived PTG led to a decrease in

(*continued on next page*)

Table 1 (*continued*)

Authors	Aims	Study Design	PTG Instrument	Recruitment period	Sample	Country	Main results related to PTG
	mental health (internalising problems, aggressive behaviour, posttraumatic stress symptoms) and health-related quality of life (HRQoL) during the first 2 years of the COVID-19 pandemic.		positive impact do you think the Corona crisis has/had on your child?"	December 2021 (t4) (during the first and the second lockdown).		and South Tyrol (Italy)	internalizing problems, aggressive behavior, and posttraumatic symptoms as well as increased HRQoL in older children. Parents at t4 reported less PTG for children than parents at t1. PTG at t4 did not differ between younger and older children. Though almost every child (97.6 % of younger, 85.7 % of older) showed signs of PTG at t1, almost three-quarters of the children (71.9 % of younger, 71.7 % of older) showed signs of PTG at t4.
Zeng et al. (2022)	This study investigated the characteristics of PTG and academic burnout among adolescents in an ethnic minority area in China, and the moderating role of core belief challenge on the association between PTG and academic burnout.	Cross-sectional survey	The Posttraumatic Growth Inventory	From April 29, 2020 to May 13, 2020 (during the first lockdown).	Adolescents (12–18 years old)	China	In this study, 32.3 % of the adolescents reported moderate or high levels of PTG. PTG among junior high school students was significantly higher than that of senior high school students. PTG had a significant negative predictive effect on academic burnout, while core belief challenge did not. However, the interaction between PTG and core belief challenge had a significant negative predictive effect on academic burnout.
Xu et al. (2023)	This study examined the post-traumatic reactions (i.e., PTSD, depression, anxiety) and PTG after trauma and the influence of family function on post-traumatic reactions	Cross-sectional survey	The Posttraumatic Growth Inventory	From December 10 to December 15, 2022 (after the second lockdown).	Adolescents (11–18 years old)	China	In this study, 47 % of adolescents infected with COVID-19 had low PTSD, depression, and anxiety and high PTG scores. Various dimensions of family function such as problem-solving, behavioral control, roles in family function, and general family function were protective factors for adolescents to develop PTG and cope with PTSD, depression, and anxiety.
Zhang et al. (2021)	This study explored (1) mental health status in discharged children who had COVID-19 compared with healthy children who were quarantined at home and (2) risk factors that were associated with mental health problems in discharged children.	Case control design	The Post-traumatic Growth Inventory–Short Form	From July to September 2020 (after the first COVID-19 lockdown).	Discharged Children (7–12 years old)	China	PTG was higher in primary school children and children who were more resilient.
Zhen and Zhou (2022)	This study examined co-occurring patterns of PTSD symptoms, depressive symptoms, and PTG among adolescents.	Cross-sectional survey	The Posttraumatic Growth Inventory	July 2020 (after the first COVID-19 lockdown).	Adolescents (15–18 years old)	China	The prevalence of PTG was 22 %. PTG coexisted with PTSD. Positive refocusing and reappraisal were associated with high PTG and low PTSD and depressive symptoms.
Zhen et al. (2022)	This study examined mechanisms underlying the effects of parent-child	Cross-sectional survey	The Posttraumatic Growth Inventory	July 2020 (after the first COVID-19 lockdown).	Adolescents (15–18 years old)	China	The prevalence of PTG was 22.4 % and the mean score was 49.35. Open

(*continued on next page*)

Table 1 (*continued*)

Authors	Aims	Study Design	PTG Instrument	Recruitment period	Sample	Country	Main results related to PTG
	communication on PTSD and PTG in adolescents						parent-child communication, self-compassion and self-disclosure were positively associated with PTG whereas problematic parent-child communication was negatively associated with PTG.
Zhou et al. (2021)	This study aimed to (a) describe the prevalence of depression, anxiety, and PTG among adolescents (b) explore potential challenges and opportunities for adolescents; and (c) investigate the risk and protective factors of depression, anxiety, and PTG.	Cross-sectional survey	The Posttraumatic Growth Inventory	From February 23 to March 8, 2020 (during the first lockdown).	Adolescents (16–26 years old)	China	The prevalence of PTG was 45.6 %. Female sex and older age were associated with higher levels of PTG. Symptoms related to COVID-19, excessive attention to epidemic information, living in urban or severe epidemic areas, poor family economic status, and conflicts with parents during home quarantine were risk factors for PTG. Frequent communication with peers, exercise, and receiving public welfare psychological assistance were protective factors.

study design, the PTG instrument, the recruitment period, the number and age of the participants, the country of reference, and a short summary of the findings with reference to the PTG prevalence rates and contributing factors . A synthesis of the evidence followed.

3. Results

3.1. Search strategy and outcomes

The initial search yielded a total of 136 articles. Database-specific results were as follows: PubMed: 36 articles retrieved using advanced search filters and MeSH terms; Scopus: 45 articles identified using keyword combinations and publication filters; PsycINFO: 22 articles retrieved using the Thesaurus of Psychological Index Terms and Boolean operators; Google Scholar: Due to the broad and less-specific indexing of this database, a targeted manual screening of the first 84 results sorted by relevance was conducted, yielding 33 potentially relevant articles.

After removing 21 duplicates, 115 sarticles were then assessed by reading the title and abstract for relevance based on the inclusion criteria: (1) focus on PTG (2) examined PTG among children and adolescents (3) published during or after the COVID-19 pandemic, (4) peer-reviewed empirical studies, dissertations/theses and (5) published in English. Of those reviewed, 80 articles were excluded, leaving a total of 33 articles that were assessed for eligibility. After full-text reading, 12 articles were excluded because the sample was not children or adolescents. Eventually, a total of 21 articles were considered relevant to the review. The data abstraction process is shown in Fig. 1.

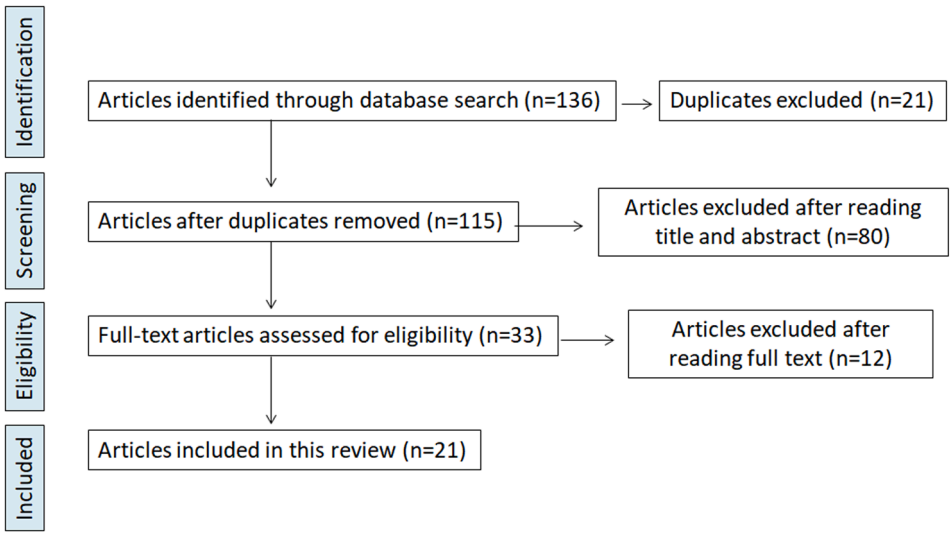

Fig. 1. Flow diagram of the literature search and article selection.

3.2. Posttraumatic growth in different COVID-19 periods and countries

Excluding one study that did not report a specific recruitment period (Kim et al., 2022), the reamaing studies were conducted during different recruitment periods across the pandemic; seven during the first lockdown (Jian et al., 2022; Kohlboeck et al., 2023; Senejko et al., 2022; Ulset & von Soest, 2022; Zeng et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021), three after the first lockdown (Waters et al., 2020; Zhang et al., 2021; Zhen et al., 2022), three during the second lockdown (Bhushan et al., 2022; Kazemi et al., 2022; Lanfredi et al., 2023) and two after the second lockdown (Casali et al., 2023; Xu et al., 2023). Among the longitudinal studies, Tang et al. (2022) examined the PTG during and after the first lockdown, Wenter et al. (2022) examined the perceived PTG during the first and second lockdowns, and Taddei et al. (2024) examined PTG at three time points: the first in January/February 2020 (T0) before the first COVID-19 outbreak, the second in May/June 2020 (T1) after the COVID-19 lockdown, the third in May/June 2021 (T2), and the last in May/June 2022 (T3). The experimental study by Tamiolaki et al. (2024) examined PTG at three timepoints after the second lockdown (pre-post intervention and follow-up). Hamama et al. (2024) examined PTG two years after the COVID-19 outbreak.

The vast majority of the studies were conducted in China (Jian et al., 2022; Tang et al., 2022; Xu et al., 2023; Zeng et al., 2022; Zhang et al., 2021; Zhen et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021), whereas the remaining were conducted in Italy (Casali et al., 2023; Hamama et al., 2024; Lanfredi et al., 2023), Korea (Kim et al., 2022), Australia (Waters et al., 2020), Norway (Ulset & von Soest, 2022) North Tyrol (Austria) and South Tyrol (Italy) (Kohlboeck et al., 2023; Wenter et al., 2022), India (Bhushan et al., 2022), Poland (Senejko et al., 2022), Iran (Kazemi et al., 2022), Israel (Taddei et al., 2024) and Greece (Tamiolaki et al., 2024) (Fig. 2).

3.3. Study design

The majority of the studies were cross-sectional (Bhushan et al., 2022; Hamama et al., 2024; Jian et al., 2022; Kazemi et al., 2022; Lanfredi et al., 2023; Senejko et al., 2022; Ulset & von Soest, 2022; Xu et al., 2023; Zeng et al., 2022; Zhang et al., 2021; Zhen et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021), with the exception of four longitudinal studies (Casali et al., 2023; Tang et al., 2022; Taddei et al., 2024; Wenter et al., 2022), two experimental studies (Kim et al., 2022; Tamiolaki et al., 2024) and one case control study (Zhang et al., 2021).

3.4. Posttraumatic growth measures

Eleven studies (Hamama et al., 2024; Jian et al., 2022; Kazemi et al., 2022; Kim et al., 2022; Senejko et al., 2022; Tang et al., 2022; Xu et al., 2023; Zeng et al., 2022; Zhen et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021) used the Post Traumatic Growth Inventory (PTGI) (Tedeschi & Calhoun, 1996), four studies (Bhushan et al., 2022; Casali et al., 2023; Taddei et al., 2024; Tamiolaki et al., 2024) used the Revised Post-traumatic Growth Inventory for Children (PTGI-C-R), three studies (Lanfredi et al., 2023; Ulset & von Soest, 2022; Zhang et al., 2021) used the Post-traumatic Growth Inventory–Short Form (PTGI-SF), two other studies used the open-ended question "What positive impact do you think the coronavirus crisis has/had on your child?" (Kohlboeck et al., 2023; Wenter et al., 2022), and one study used the Stress-Related Growth Scale (Waters et al., 2020).

3.5. Prevalence rates or levels of posttraumatic growth

Nearly all studies reported exact prevalence rates and/or general levels of PTG (Bhushan et al., 2022; Casali et al., 2023; Jian et al., 2022; Kim et al., 2022; Lanfredi et al., 2023; Taddei et al., 2024; Tang et al., 2022; Tamiolaki et al., 2024; Ulset & von Soest, 2022; Waters et al., 2020; Wenter et al., 2022; Xu et al., 2023; Zeng et al., 2022; Zhen et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021). Diverse levels and prevalence rates of PTG have been reported depending on the country, the recruitment period, the sample, and the assessment instrument. For example, a PTGI cut-off score below 45 indicates no or low PTG, above 46 indicates moderate to high PTG, whereas PTG-C-R cut-off score below 19 suggests non or low PTG, and above 20 suggests moderate to high PTG.

During the first lockdown, a study in Norway (Ulset & von Soest,

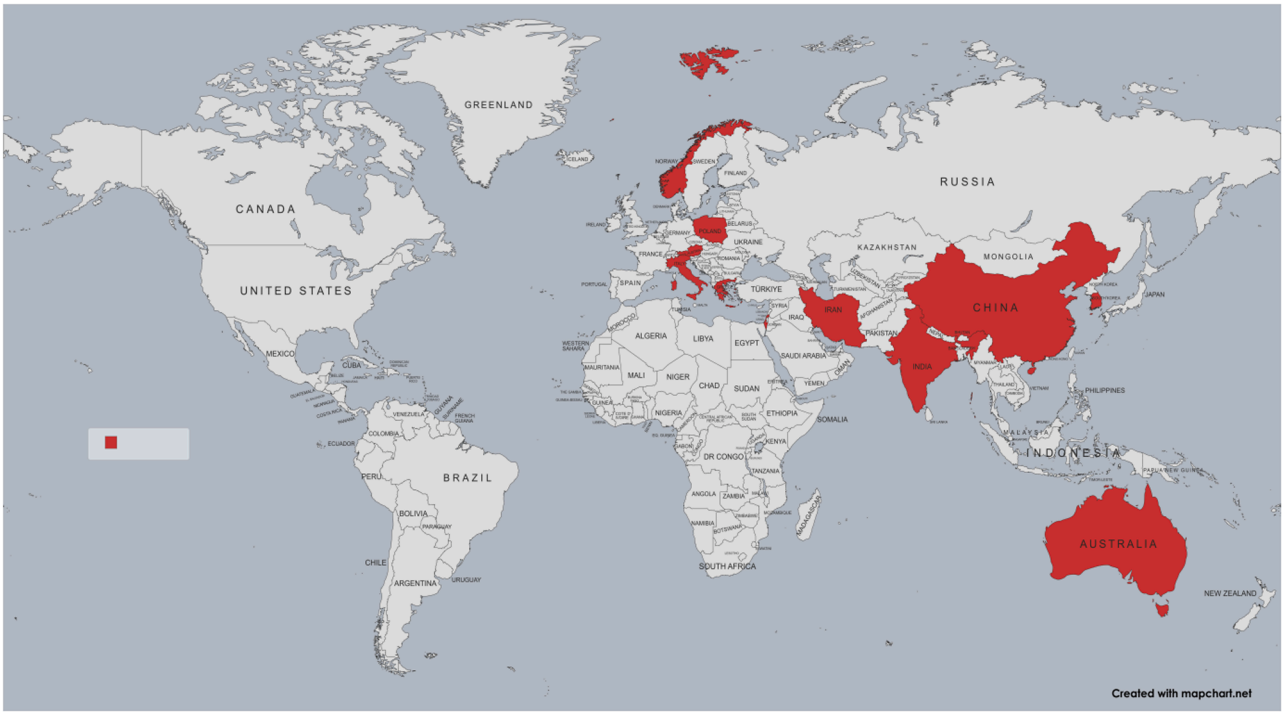

Fig. 2. PTG research map among children and adolescents in different countries.

2022) revealed that 9.6 % of the adolescents experienced moderate to high levels of PTG, whereas a study in Italy (Wenter et al., 2022) showed that almost every child (97.6 % of younger, 85.7 % of older children) showed signs of PTG. In the same period in China, 20.6 % (Jian et al., 2022) or 45.6 % (Zhou et al., 2021) of the adolescents reported PTG and 32.3 % (Zeng et al., 2022) experienced moderate or high levels of PTG. In the longitudinal study in China by Tang et al. (2022), 22.4 % of adolescents experienced PTG during the first lockdown, which increased to 25.0 % after the first lockdown. A similar percentage of moderate to high levels of PTG (22.4 %) was reported after the first lockdown in China (Zhen et al., 2022).

During the second lockdown, almost three-quarters of the children in Italy (71.9 % of younger, and 71.7 % of older children) showed signs of PTG (Wenter et al., 2022). In an Indian sample of children and adolescents, Bhushan et al. (2022) found that 39.8 % experienced PTG (without reporting differences between children and adolescents). After the second lockdown, Casali et al. (2023) found moderate to high levels of PTG among adolescents,and Xu et al. (2023) showed that 47 % of the adolescents infected with COVID-19 experienced high levels of PTG, whereas Lanferdi et al. (2022) found that the adolescents experienced low levels of PTG. In Kim et al. (2022) videoconferencing-based home exercise program among childhood cancer survivors in Korea, the levels of PTG were moderate to high at baseline and did not significantly improve after the intervention. After the second lockdown, in Tamiolaki et al's (2024) mindfulness-character strength intervention study among children in Greece, levels of PTG were low at baseline, and moderate to high both after the intervention and at the follow-up. In the study by Taddei et al. (2024), a significant linear decline in PTG was foubd over the two years of the pandemic.

3.6. Posttraumatic growth and target group

The majority of the studies examined posttraumatic growth in adolescents (Casali et al., 2023; Hamama et al., 2024; Jian et al., 2022; Kazemi et al., 2022; Lanfredi et al., 2023; Senejko et al., 2022; Taddei et al., 2024; Tang et al., 2022; Ulset & von Soest, 2022; Waters et al., 2020; Xu et al., 2023; Zeng et al., 2022; Zhen et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021), only four studies among children (Kim et al., 2022; Kohlboeck et al., 2023; Tamiolaki et al., 2024; Wenter et al., 2022) and two studies among both children and adolescents (Bhushan et al., 2022; Zhang et al., 2021).

3.7. Posttraumatic growth across age and gender

Most researchers reported that older age was associated with lower PTG (Zeng et al., 2022; Zhang et al., 2021; Zhou et al., 2021), whereas two studies found that older age was associated with higher PTG (Tamiolaki et al., 2024; Zhen & Zhou, 2022). The seemingly contradictory findings may be interpreted in terms of the age range of the samples. For example, the study byTamiolaki et al. (2024) included children aged 8–10 years, whereas the study by Zhang et al. (2021) included children aged 7–12 years. Ten studies examined gender differences. During the COVID-19 pandemic, some researchers found that males experienced higher PTG than females (Casali et al., 2023; Jian et al., 2022; Kazemi et al., 2022; Tang et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021), whereas others found the opposite (Lanfredi et al., 2023; Senejko et al., 2022; Tamiolaki et al., 2024; Ulset & von Soest, 2022).

3.8. Posttraumatic growth and COVID-19-related factors

The COVID-19 experience seemed to play a key role in PTG. However, some of the findings are inconclusive. Kazemi et al., 2022 showed that adolescents without COVID-19 were more likely to achieve PTG, whereas Xu et al. (2023) showed that 47 % of the adolescents with COVID-19 experienced high PTG. The COVID-19 threats to family, life, and lifestyle were found either to activate PTG (Senejko et al., 2022) or were shown to be risk factors for PTG (Zhou et al., 2021). Concerns about the pandemic, being homeschooled adolescents, immigrant, having low neighborhood socioeconomic status, low income, parents working remotely (Kazemi et al., 2022; Ulset & von Soest, 2022) and greater exposure to electronic media (Bhushan et al., 2022) increased the likelihood of PTG.

3.9. Posttraumatic growth and individual factors

Different coping strategies and personal traits were significantly associated with PTG. Researchers (Waters et al., 2020; Zhen & Zhou, 2022) found that positive refocusing and reappraisal helped adolescents reframe the COVID-19 experience, find new meaning, and make positive changes in their lives. Waters et al., 2020 also found that emotional processing, positive reappraisal, and strengths use were mediators between positive education and growth. Zeng et al. (2022) showed that a higher core belief challenge can help adolescents rethink the value of life, experience positive changes, and resist the risk of increased academic burnout. Tamiolaki et al. (2024) demonstrated that strength use and mindfulness were predictors of PTG. Casali et al. (2023) found that perseverance was related to PTG; students with perseverance were able to transform the COVID-19 crisis by developing a new outlook that enabled PTG and helped them to be more satisfied with their lives at the end of the school year. Senejko et al. (2022) found extroversion to be a predictor of PTG and suggested that the pandemic may motivate extroverts to reflect and reconsider their relationships with others, which is an important aspect of PTG. Additionally, emotional intelligence was found to significantly predict higher PTG (Tang et al., 2022). The researchers suggested that people with high emotional intelligence may be more likely to use adaptive (rational, intuitive) rather than maladaptive (dependent, avoidant) cognitive styles to cope with stress, or that emotional intelligence may contribute to resilience, which in turn may enhance personal strength which is a subcomponent of PTG. Furthermore, the same study (Tang et al., 2022) argued that perceived emotional intelligence predicted adolescents' self-esteem and promoted PTG. Adolescents with higher emotional intelligence and self-esteem (perceptions of self-worth and value) seemed to evaluate the COVID-19 experience in a more positive way and experienced more positive changes (Tang et al., 2022). Also, emotional regulation played a mediating role in the relationship between self-esteem and PTG (Tang et al., 2022); participants with higher emotional intelligence were better able to regulate their emotions and consider COVID-19 difficulties as opportunities for PTG. In another study, lower levels of self-efficacy reduced the likelihood of PTG (Jian et al., 2022). It seems that high self-efficacy provides internal and external resources that promote growth and positive psychological transformations. High levels of self-compassion and self-disclosure were also positively associated with PTG (Zhen et al., 2022); self-compassion and self-disclosure seem to help adolescents move beyond their original cognitive appraisals to establish new core beliefs, thus promoting PTG development.

3.10. Posttraumatic growth and relational factors

With regard to interpersonal variables, social support has been extensively studied. According to researchers (Bhushan et al., 2022; Kazemi et al., 2022; Ulset & von Soest, 2022; Zhen et al., 2022; Zhou et al., 2021), PTG has been positively associated with social support (i.e., peer support, support from supervisors, and support from friends and family). Specifically, good relationships with parents and peers, parental care, having someone to talk to, and receiving public welfare or psychological assistance were positive predictors of PTG (Ulset & von Soest, 2022; Zhou et al., 2021). Zhen et al. (2022) demonstrated that when parents and adolescents communicate openly, adolescents have greater perceived parental emotional support, which among other variables, promotes PTG. Similarly, Ulset and von Soest (2022) found that caring

parents who were willing to discuss the pandemic with their offspring helped them to engage in more deliberate and constructive thinking about the pandemic, which, in turn, fostered PTG in their offspring. Hamama et al. (2024) found that higher levels of PTG in parents were also associated with higher levels of PTG in adolescents. On the other hand, problematic parent-child communication and conflicts with parents have been identified as risk factors that have been associated with less PTG (Zhen et al., 2022; Zhou et al., 2021). Moreover, it has been shown that children and adolescents with at least one or more siblings experienced higher PTG (Bhushan et al., 2022; Tamiolaki et al., 2024). Furthermore, Xu et al. (2023) found that various dimensions of family functioning, such as problem-solving (i.e., identifying problems, thinking about various solutions to problems, selecting and implementing appropriate solutions, and evaluating the effects of solutions), behavioral control (i.e., restricting family members' behavior when a family responds to various environmental pressures, including risky behaviors, basic physiological needs and family members' communication), roles in family function (i.e., the relative position of family members in the family, the responsibilities of family members, and the behavior pattern of family members) and general family functioning were protective factors of PTG.

3.11. Posttraumatic growth and psychological factors / mental health

A few studies (Kazemi et al., 2022; Tamiolaki et al., 2024; Ulset & von Soest, 2022) have examined the psychological protective and risk factors for PTG. Higher levels of satisfaction with family, with school systems (Taddei et al., 2024), with life and well-being were associated with a higher likelihood of experiencing PTG (Kazemi et al., 2022; Tamiolaki et al., 2024; Ulset & von Soest, 2022), whereas lower levels of resilience and life satisfaction were risk factors for PTG (Ulset & von Soest, 2022; Zhang et al., 2021). Stress (Senejko et al., 2022; Taddei et al., 2024) and PTSD (Kohlboeck et al., 2023; Tamiolaki et al., 2024) were also found to promote PTG. Low levels of PTSD (Kohlboeck et al., 2023; Tamiolaki et al., 2024) and medium levels of stress have been associated with higher levels of PTG among adolescents (Senejko et al., 2022; Taddei et al., 2024), whereas high levels of PTSD, depression, and anxiety have been associated with lower levels of PTG (Xu et al., 2023). Jian et al. (2022) found a reverse U-shaped curvilinear relationship between PTSD and PTG in adolescents, and self-efficacy was positively correlated with PTG. Jian et al. (2022) reported that for adolescents with low self-efficacy, an increase in their PTSD symptoms aligned with an increase in PTG, whereas for adolescents with high self-efficacy, an increase in their PTSD symptoms resulted in a slow decline.

Other studies have examined factors that co-occur with PTG. Researchers have shown that PTG as a positive psychological reaction coexisted with negative psychological responses, such as depression, anxiety in adolescents (Zhou et al., 2021), post-traumatic stress in children and adolescents (Bhushan et al., 2022; Zhen & Zhou, 2022) and COVID-19 related stress in adolescents (Taddei et al., 2024). These findings suggest that PTG, depression, anxiety, stress and PTSD should be considered as outcomes of separate processes. Some studies have also examined the mental health benefits of PTG. In the study by Wenter et al. (2022), perceived PTG increased health-related quality of life and reduced internalizing problems, posttraumatic stress symptoms, and aggressive behavior in older children, but not in younger ones. In addition, Zeng et al. (2022) found that PTG had a significant negative predictive effect on academic burnout; the higher the level of adolescents' PTG, the lower the level of academic burnout.

4. Discussion

The current pandemic offered both challenges and opportunities to children and adolescents (Zhou et al., 2021). The concept of PTG has been particularly attractive to many researchers as it suggests that psychopathology such as posttraumatic stress disorder is not the only direct outcome of adversity, but people can derive certain positive outcomes from the struggle with life-threatening experiences. To the best of our knowledge, this review represents the first attempt to comprehensively synthesize the literature on COVID-19-related PTG rates and related factors among children and adolescents.

Interestingly, despite the proliferation of studies on adult PTG, there is a paucity of research in children and adolescents, and this is particularly true for children (Kim et al., 2022; Kohlboeck et al., 2023; Tamiolaki et al., 2024; Wenter et al., 2022) rather than adolescents (Casali et al., 2023; Hamama et al., 2024; Jian et al., 2022; Kazemi et al., 2022; Lanfredi et al., 2023; Senejko et al., 2022; Taddei et al., 2024; Tang et al., 2022; Ulset & von Soest, 2022; Waters et al., 2020; Xu et al., 2023; Zeng et al., 2022; Zhen et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021). According to Tedeschi & Calhoon (2004), children are less likely to have the cognitive capacity to experience PTG compared to adolescents and young adults who have an established set of schemas that change after the trauma. Nonetheless, studies have shown that PTG is possible as early as seven years of age (Kilmer, 2006).

Not surprisingly, more research on PTG has been conducted in China during the first lockdown (Jian et al., 2022; Tang et al., 2022; Xu et al., 2023; Zeng et al., 2022; Zhang et al., 2021; Zhen et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021). Diverse levels and prevalence rates of PTG have been reported. Most researchers have provided PTG levels ranging from moderate to high (Casali et al., 2023; Ulset & von Soest, 2022; Zeng et al., 2022), whereas others have provided low rates (Lanferdi et al., 2022; Tamiolaki et al., 2024), thus making the picture unclear. Reasons for these discrepancies may be the different countries and associated cultural backgrounds, the recruitment periods, and the diverse instruments.

Since the majority of the studies were cross-sectional (Bhushan et al., 2022; Hamama et al., 2024; Jian et al., 2022; Kazemi et al., 2022; Lanfredi et al., 2023; Senejko et al., 2022; Ulset & von Soest, 2022; Xu et al., 2023; Zeng et al., 2022; Zhang et al., 2021; Zhen et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021), they were inconclusive as to whether PTG levels pre-existed or to what extent increased because of the adverse experience of the pandemic. Nor do the longitudinal studies (Casali et al., 2023; Tang et al., 2022; Wenter et al., 2022) allow us to draw firm conclusions about whether or how PTG has been maintained or changed over time or whether PTG represents a coping strategy, an illusion or a real growth phenomenon. The inconsistent findings range from significant to trivial increase in PTG (Tang et al., 2022; Wenter et al., 2022) and from significant to no change after the intervention (Kim et al., 2022; Tamiolaki et al., 2024).In addition, a longitudinal study (Taddei et al., 2024) suggested that the decreasing levels of PTG during the pandemic did not reflect real PTG, but rather a coping strategy to deal with the effects of COVID-19 or a positive illusion that helps adolescents maintain unrealistically optimistic beliefs in response to highly stressful or threatening situations. Examining PTG at different time points would have contributed to a more nuanced understanding of the psychological reactions to COVID-19 and more importantly, to the trajectories through which the real process of PTG occurs and would potentially guide the implementation of appropriate support interventions for children and adolescents, who are particularly vulnerable.

Gender differences were found in PTG consistent with findings in adult samples (e.g., Henson et al., 2021). Notwithstanding that the overwhelming majority of studies have found that males report higher levels of PTG than females (Casali et al., 2023; Jian et al., 2022; Kazemi et al., 2022; Tang et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021), research has not yet reached a unanimous agreement, as findings are both limited in number and inconclusive in nature. In contrast, higher levels of PTG have been rather consistently associated with younger age (Zeng et al., 2022; Zhang et al., 2021; Zhou et al., 2021) and gender (Jian et al., 2022; Tang et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021), similar to the adult literature (e.g., Bovero et al., 2023) and Tedeschi and Calhoun's theory (2004). Findings among children and adolescents in response to natural disasters are also in agreement

(Bernstein & Pfefferbaum, 2018).

A notable finding of this review was that the synthesis of studies on the facilitators of PTG was largely dispersed and inconclusive, similar to that among adults (Henson et al., 2021). A variety of COVID-19-related factors (e.g., infection by COVID-19, threats and worries to family, life, and lifestyle) (Kazemi et al., 2022; Senejko et al., 2022; Ulset & von Soest, 2022) and several diverse individual characteristics, such as different coping strategies (e.g., positive refocusing, reappraisal) (Waters et al., 2020; Zhen & Zhou, 2022) and personal traits (e.g., perseverance, extroversion, character strengths, self-compassion, self--disclosure, self-efficacy, emotional intelligence, emotional regulation) have been associated with PTG (Casali et al., 2023; Jian et al., 2022; Senejko et al., 2022; Tamiolaki et al., 2024; Tang et al., 2022; Zhen et al., 2022). Since cognitive processing is essential for PTG (Tedeschi & Calhoun, 2004), these specific individual characteristics appear to be robust facilitators of PTG. That being said, a comment should be made about the variety of facilitators that need to be taken into account to fully understand the outcome of an adversity such as the pandemic. In this regard, it would be helpful to identify individuals who are less likely to experience PTG due to their coping strategies and personality traits, and effectively support them to experience positive changes through trauma interventions .

Similar to the adult literature (Bovero et al., 2023; Henson et al., 2021; Kalaitzaki et al., 2023), social support was a consistent facilitator of PTG. Abundant studies found that material and/or emotional support received from family, friends, and the community, can be considered a key environmental resource for promoting and/or maintaining PTG in children and adolescents (Bhushan et al., 2022; Kazemi et al., 2022; Tamilaki et al. 2024; Ulset & von Soest, 2022; Xu et al., 2023; Zhen et al., 2022; Zhou et al., 2021). Social support can emphasize feelings of belonging and provide an encouraging, safe, and supportive environment, in which children and adolescents can process traumatic events and frame their experiences in a positive light (Tamiolaki et al., 2024). If social support is important for adults in fostering PTG, it could be more important for children and adolescents, as this age group is intertwined with demanding developmental tasks, such as the identity construction, critical and moral thinking, and romantic relationships (Grazzani et al., 2022). Attempts to enhance social support and belongingness seem to be fundamental for stimulating or increasing PTG.

Psychological factors, such as resilience, well-being, and satisfaction with family, school, and life were positively associated with PTG (Kazemi et al., 2022; Taddei et al., 2024; Ulset & von Soest, 2022; Zhang et al., 2021). Findings related to psychopathology indices (e.g., stress, depression, anxiety, and PTSD) were rather inconsistent, with studies finding positive (Jian et al., 2022), negative (Kohlboeck et al., 2023; Taddei et al., 2024; Xu et al., 2022), or no association with PTG (Bhushan et al., 2022; Zhen & Zhou, 2022; Zhou et al., 2021). Relevant studies in adult samples have suggested that there may be an optimal level of moderate posttraumatic stress or PTSD that strengthens PTG (Kalaitzaki, 2021; Kalaitzaki et al., 2022). Although mostly positive/adaptive coping strategies (positive refocusing and reappraisal) have been associated with PTG (Waters et al., 2020; Zhen & Zhou, 2022), suggesting its real aspect, the scant evidence cannot be conclusive as to whether PTG was real or illusory (Zoellner & Maercker, 2006). Research also needs to examine PTG's dimensions separately, as they may have different relationships with various factors; this evidence may be useful in revealing the pathways that lead to positive psychological changes.

4.1. Strengths and limitations

To the best of the authors' knowledge, this is the first review to examine studies on PTG related to COVID-19 in children and adolescents. Understanding PTG in young people provides valuable insights for developing prevention and intervention strategies that focus on growth rather than pathology. By exploring key factors such as personal, psychological, and relational traits, tailored interventions can be designed to support positive post-trauma development. Integrating PTG-based principles into clinical practice strengthens resilience, enhances protective psychosocial factors, and fosters personal strengths, leading to more effective, developmentally appropriate trauma-informed care for children and adolescents.

This review used a thorough literature search methodology (resembling that of the systematic reviews), adding value to the evidence retrieved. However, a few limitations should be acknowledged. Notwithstanding the comprehensive analysis of the literature, the studies were not assessed for their quality. Excluding the few longitudinal studies on PTG, the cross-sectional design of the majority of studies is a limitation that does not allow for causal inferences or interpretation of the associations. Only the PTG was included and potentially relevant experiences (e.g., benefit finding) were ignored. Although reporting findings from different recruitment periods across the pandemic was a strong point of this review and enlightened our understanding of PTG, it made comparisons difficult, particularly in reference to the prevalence rates. Comparisons were also difficult because of the use of diverse assessment instruments in the studies. The self-report instruments used by all studies may have also biased the findings, particularly in the samples of children. It was difficult to know whether PTG was exclusively due to COVID-19 since it may be hard for children and adolescents to disentangle other effects. Furthermore, the lack of an established standard regarding when to measure PTG affects the types of conclusions that can be drawnand the validity of the present review, although Burt and Katz (1987) suggested that some aspects of growth were independent of the passage of time. Moreover, since children and adolescents are characterized by enormous and dramatic changes and their needs may be disparate across different developmental stages (Cusinato et al., 2020), it may be useful to consider children and adolescents separately. An important limitation concerns the risk of overinterpreting cross-sectional data as indicative of real posttraumatic growth (PTG). Such designs lack temporal sensitivity and cannot capture intra-individual change or developmental trajectories.

Self-reported PTG may stem from cognitive reframing or an illusion of growth rather than actual psychological transformation. Without longitudinal validation, associations may be confounded by dispositional traits or retrospective meaning-making. Moreover, the absence of pre-trauma baseline measurements precludes assessment of true developmental shifts. Thus, cross-sectional findings should be viewed as suggestive rather than conclusive proof of real PTG. The exclusion of potentially relevant qualitative research and research published in languages other than English was another limitation of this review.

4.2. Implications for research, practice, and policy

The summarised knowledge points to numerous implications for research, practice, and policy. Considering that the current topic is still crucial and timely (e.g., potential long-term consequences of COVID-19), yet largely underexplored, future research should focus on COVID-19-related PTG in children and adolescents in the long run. To gain a clearer understanding, future studies should use longitudinal methodologies to explore how PTG unfolds across developmental stages, thus identifying key factors that influence PTG over time, including mediators and moderators that shape its progression. Longitudinal studies in different cultural contexts should immerse in the process through which PTG occurs globally, especially in underrepresented regions and create developmentally and culturally sensitive instruments to enhance the validity and generalizability of PTG findings. Future studies should also work to distinguish real PTG from its illusory counterpart, clarify their characteristics and create standardized inventories to assess attitudes towards both forms. Differentiating between real and illusory PTG is essential to prevent overestimations of psychological adaptation and ensure that interventions target meaningful, sustained change rather than superficiall coping. Furthermore, educating professionalson

these distinctionsmay reduce misinterpretations in self-reported growth. Moreover integrating mixed-methods approaches -combining standardized measures with qualitative inquiry- can enhance construct validity, uncover nuanced meaning-making processes, and capture the subjective and contextual dimensions of PTG. Such methodological refinement is essential for advancing theoretical clarity and practical applications in clinical settings.

Although widely heterogeneous, the modifiable personal and environmental resources and protective factors that were identified to facilitate PTG can be used by public health authorities, policymakers, and clinicians to develop and implement interventions to promote PTG targeted at the vulnerable population subgroup of children and adolescents. Such interventions may include restoring or boosting mental health through (group) counseling, meditation, (self-)relaxation, and other techniques, promoting positive personal traits (e.g., character strengths), teaching adaptive coping strategies, and enhancing social support systems through providing opportunities for positive social interactions. Given the protracted pandemicand the evolving needs of children and adolescents, evaluations should be regular and interventions should be ongoing. In this respect, interventions might be effective in meeting both short-term needs, such as providing relief and safeguarding mental health in times of intense crises, and long-term evolving needs, such as psychologically empowering children and adolescents and ameliorating their well-being and quality of life.

5. Conclusion

Recognizing PTG as a public mental health resource holds strategic value, particularly in the context of pandemics or large-scale crises. Integrating PTG into mental health strategies shifts the focus from problems and deficits to strengths-based approaches that promote positive psychological growth and fosters collective recovery processes in the aftermath of shared adversity. This scoping review first revealed that, notwithstanding the importance of PTG, years after the outburst of the pandemic, there is a profound dearth of research among COVID-19-exposed children and adolescents. The synthesis of knowledge provided insights into the heterogeneous rates and associated factors of PTG induced by COVID-19. Importantly, PTG was feasible in children and adolescents as a result of the pandemic, similar to adults. The findings identified notable, though fairly dispersed, PTG key predictors, including COVID-19-related factors, sociodemographics, coping strategies, mental health, personality, and psychological traits, with social support being the outstanding and consistent facilitator. Based on the findings, schools should create psychologically safe environments that enable students to process adversity constructively. Clinicians may apply PTG principles to promote posttraumatic change beyond symptom relief. Policymakers should embed growth-oriented, developmentally informed approaches into youth mental health policy to strengthen public health crisis preparedness and long-term recovery capacity. Many gaps were also identified, including the need for culturally informed longitudinal studies to track PTG, distinguish real from illusory PTG, investigate PTG in underrepresented regions, and develop culturally sensitive and age-appropriate instruments for children and adolescents.

Ethical approval

It was not required as this study reviewed previously published data.

Funding

This research did not receive any specific grand from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Alexandra Tamiolaki: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Argyroula Kalaitzaki:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Abbas, S., Abbas, H. S. M., & Zhang, Y. (2024). Psychological implications of man-made trauma and efficacy of "Teaching Recovery Techniques" on children and adolescents exposed to trauma. *European Journal of Trauma & Dissociation, 8*(2), Article 100397. https://doi.org/10.1016/j.ejtd.2024.100397

Amjadi, K. (2021). Exploring factors that influence children's growth and development during a pandemic. *Global Pediatric Health, 8*. https://doi.org/10.1177/2333794X211042464, 2333794×2110424.

Araújo, L. A.de, Veloso, C. F., Souza, M.de C., Azevedo, J. M. C.de, & Tarro, G. (2021). The potential impact of the COVID-19 pandemic on child growth and development: A systematic review. *Journal de Pediatria, 97*(4), 369–377. https://doi.org/10.1016/j.jped.2020.08.008

Arksey, H., & O'Malley, L. (2005). Scoping studies: Towards a methodological framework. *International Journal of Social Research Methodology, 8*(1), 19–32. https://doi.org/10.1080/1364557032000119616

Bernstein, M., & Pfefferbaum, B. (2018). Posttraumatic growth as a response to natural disasters in children and adolescents. *Current Psychiatry Reports, 20*(5). https://doi.org/10.1007/s11920-018-0900-4

Bhushan, B., Basu, S., & Ganai, U. J. (2022). Post-traumatic stress and growth among the children and adolescents in the aftermath of COVID-19. *Frontiers in Psychology, 12*. https://doi.org/10.3389/fpsyg.2021.791263

Bovero, A., Balzani, S., Tormen, G., Malandrone, F., & Carletto, S. (2023). Factors associated with post-traumatic growth during the COVID-19 Pandemic: A systematic review. *Journal of Clinical Medicine, 13*(1), 95. https://doi.org/10.3390/jcm13010095

Burt, M. R., & Katz, B. L. (1987). Dimensions of recovery from rape. *Journal of Interpersonal Violence, 2*(1), 57–81. https://doi.org/10.1177/088626087002001004

Casali, N., Feraco, T., & Meneghetti, C. (2023). Keep going, keep growing: A longitudinal analysis of grit, posttraumatic growth, and life satisfaction in school students under COVID-19. *Learning and Individual Differences, 105*, Article 102320. https://doi.org/10.1016/j.lindif.2023.102320

Compas, B. E., Connor-Smith, J. K., Saltzman, H., Thomsen, A. H., & Wadsworth, M. E. (2001). Coping with stress during childhood and adolescence: Problems, progress, and potential in theory and research. *Psychological Bulletin, 127*(1), 87–127. https://doi.org/10.1037//0033-2909.127.1.87

Cusinato, M., Iannattone, S., Spoto, A., Poli, M., Moretti, C., Gatta, M., & Miscioscia, M. (2020). Stress, Resilience, and Well-Being in Italian Children and Their Parents during the COVID-19 Pandemic. *Int J Environ Res Public Health., 17*(22), 8297. https://doi.org/10.3390/ijerph17228297. Nov 10.

Domen, I. (2024). Narrative exposure therapy for PTSD and structural dissociation and techniques for its conduct: A case study in Japanese psychiatric hospital. *European Journal of Trauma & Dissociation, 8*(2), Article 100415. https://doi.org/10.1016/j.ejtd.2024.100415

Fayaz, I. (2023). Posttraumatic growth and trauma from natural disaster in children and adolescents: - A systematic literature review about related factors. *Journal of Aggression, Maltreatment & Trauma, 32*(3), 305–324. https://doi.org/10.1080/10926771.2022.2133656

Ferris, C., & O'Brien, K. (2022). The ins and outs of posttraumatic growth in children and adolescents: A systematic review of factors that matter. *Journal of Traumatic Stress, 35*(5), 1305–1317. https://doi.org/10.1002/jts.22845

Frazier, P., Conlon, A., & Glaser, T. (2001). Positive and negative life changes following sexual assault. *Journal of Consulting and Clinical Psychology, 69*(6), 1048–1055. https://doi.org/10.1037//0022-006x.69.6.1048

Fung, H. W., Lam, C. C., Yuan, G. F., Chan, C. T. L., Liu, C., & Wang, E. K. S. (2024). Is childhood trauma screening distressing? Preliminary findings from two culturally different samples of young people. *European Journal of Trauma & Dissociation, 8*(1), Article 100393. https://doi.org/10.1016/j.ejtd.2024.100393

Grazzani, I., Agliati, A., Cavioni, V., Conte, E., Gandellini, S., Lupica Spagnolo, M., Ornaghi, V., Rossi, F. M., Cefai, C., Bartolo, P., Camilleri, L., & Oriordan, M. R. (2022). Adolescents' Resilience during COVID-19 Pandemic and its mediating role in the association between SEL skills and Mental health. *Frontiers in Psychology, 13*, Article 801761. https://doi.org/10.3389/fpsyg.2022.801761

Grubaugh, A. L., & Resick, P. A. (2007). Posttraumatic growth in treatment-seeking female assault victims. *Psychiatric Quarterly, 78*(2), 145–155. https://doi.org/10.1007/s11126-006-9034-7

Hamama, L., Finklestein, M., & Hamama-Raz, Y. (2024). Adolescents' posttraumatic growth during the COVID-19 pandemic: The links between differentiation of self, parents' posttraumatic growth, and adolescents' gender. *Children and Youth Services Review, 167*, Article 108022. https://doi.org/10.1016/j.childyouth.2024.108022

Harmon, J., & Venta, A. (2021). Adolescent posttraumatic growth: A review. *Child Psychiatry and Human Development, 52*(4), 596–608. https://doi.org/10.1007/s10578-020-01047-9

Henson, C., Truchot, D., & Canevello, A. (2021). What promotes post traumatic growth? A systematic review. *European Journal of Trauma & Dissociation, 5*(4). https://doi.org/10.1016/j.ejtd.2020.100195. Article 100195.

Jian, Y., Hu, T., Zong, Y., & Tang, W. (2022). Relationship between post-traumatic disorder and posttraumatic growth in COVID-19 home-confined adolescents: The moderating role of self-efficacy. *Current Psychology*. https://doi.org/10.1007/s12144-021-02515-8

Kalaitzaki, A., Tamiolaki, A., & Tsouvelas, G. (2021). From secondary traumatic stress to vicarious posttraumatic growth amid COVID-19 lockdown in Greece: The role of health care workers' coping strategies. *Psychological Trauma: Theory, Research, Practice and Policy, 14*(2), 273–280. https://doi.org/10.1037/tra0001078

Kalaitzaki, A., Tsouvelas, G., Tamiolaki, A., & Konstantakopoulos, G. (2022). Post-traumatic stress symptoms during the first and second COVID-19 lockdown in Greece: Rates, risk, and protective factors. *International Journal of Mental Health Nursing*. https://doi.org/10.1111/inm.12945

Kalaitzaki, A., Tsouvelas, G., & Tamiolaki, A. (2023). Perceived posttraumatic growth and its psychosocial predictors during two consecutive COVID-19 lockdowns. *International Journal of Stress Management, 30*(3), 223–234. https://doi.org/10.1037/str0000273

Kazemi, K., Esfahani, H., Amiri, H., Tavan, A., & Farahmandnia, H. (2022). Post-traumatic growth and perceived social support in young adolescents during the COVID-19 pandemic. *Journal of Emergency Practice and Trauma, 9*(1), 38–43. https://doi.org/10.34172/jept.2022.27

Kilmer, R. P. (2006). Resilience and posttraumatic growth in children. In L. G. Calhoun, & R. G. Tedeschi (Eds.), *Handbook of posttraumatic growth: Research and practice* (pp. 264–288). Lawrence Erlbaum Associates, Publishers.

Kim, Y., Chae, H., & Park, S. J. (2022). Feasibility and benefits of a videoconferencing-based home exercise programme for paediatric cancer survivors during the coronavirus disease 2019 pandemic. *European Journal of Cancer Care, 31*(5). https://doi.org/10.1111/ecc.13624. Portico.

Kleim, B., & Ehlers, A. (2009). Evidence for a curvilinear relationship between posttraumatic growth and posttrauma depression and PTSD in assault survivors. *Journal of Traumatic Stress, 22*(1), 45–52. https://doi.org/10.1002/jts.20378. Portico.

Kohlboeck, G., Wenter, A., Sevecke, K., & Exenberger, S. (2023). Differences in perceived threat and trauma in children during the COVID-19 pandemic. *Child and Adolescent Psychiatry and Mental Health, 17*(1). https://doi.org/10.1186/s13034-023-00628-5

Koutná, V., & Blatný, M. (2020). Socialization of coping in pediatric oncology settings: Theoretical consideration on parent-child connections in posttraumatic growth. *Frontiers in Psychology, 11*, Article 554325. https://doi.org/10.3389/fpsyg.2020.554325

Lanfredi, M., Dagani, J., Geviti, A., Di Cosimo, F., Bussolati, M., Rillosi, L., Albini, D., Pizzi, M., Ghidoni, R., Fazzi, E., Vita, A., & Rossi, R. (2023). Risk and protective factors associated with mental health status in an Italian sample of students during the fourth wave of COVID-19 pandemic. *Child and Adolescent Psychiatry and Mental Health, 17*(1). https://doi.org/10.1186/s13034-023-00615-w

Lee, H., Noh, Y., Seo, J. Y., Park, S. H., Kim, M. H., & Won, S. (2021). Impact of the COVID-19 pandemic on the mental health of adolescent students in Daegu, Korea. *Journal of Korean Medical Science, 36*(46). https://doi.org/10.3346/jkms.2021.36.e321

Li, Q., Zhu, Y., Qi, X., Lu, H., Han, N., Xiang, Y., Guo, J., & Wang, L. (2024). Posttraumatic growth of medical staff during COVID-19 pandemic: A scoping review. *BMC Public Health, 24*(1), 460. https://doi.org/10.1186/s12889-023-17591-7

Linley, P. A., & Joseph, S. (2004). Positive change following trauma and adversity: A review. *Journal of Traumatic Stress, 17*(1), 11–21. https://doi.org/10.1023/b:Jots.0000014671.27856.7e. Portico.

Loades, M. E., Chatburn, E., Higson-Sweeney, N., Reynolds, S., Shafran, R., Brigden, A., … Crawley, E. (2020). Rapid systematic review: The impact of social isolation and loneliness on the mental health of children and adolescents in the context of COVID-19. *Journal of the American Academy of Child & Adolescent Psychiatry, 59*(11). https://doi.org/10.1016/j.jaac.2020.05.009, 1218–1239.e3.

Ma, L., Mazidi, M., Li, K., Li, Y., Chen, S., Kirwan, R., … Wang, Y. (2021). Prevalence of mental health problems among children and adolescents during the COVID-19 pandemic: A systematic review and meta-analysis. *Journal of Affective Disorders, 293*, 78–89. https://doi.org/10.1016/j.jad.2021.06.021

Madrigal Gonzaleza, I. J., González-Castroa, T. B., Tovilla-Zárate, C. A., Hernández-Díaz, Y., Juárez-Rojop, I. E., Hernández-Vicencio, J. L., López-Narváez, M. L., Uresti-Rivera, E. E., Rebollo-Meza, J., Frias de la Cruz, M., & del, C. (2024). Adverse childhood experiences, perceived stress, and chronic diseases among older adults: A cross-sectional study in Mexico. *European Journal of Trauma & Dissociation, 8*(2), Article 100412. https://doi.org/10.1016/j.ejtd.2024.100412

Masten, A. S., & Motti-Stefanidi, F. (2020). Multisystem resilience for children and youth in disaster: Reflections in the context of COVID-19. *Adversity and Resilience Science, 1*(2), 95–106. https://doi.org/10.1007/s42844-020-00010-w

Mohler-Kuo, M., Dzemaili, S., Foster, S., Werlen, L., & Walitza, S. (2021). Stress and mental health among children/adolescents, their parents, and young adults during the first COVID-19 lockdown in Switzerland. *International Journal of Environmental Research and Public Health, 18*(9), 4668. https://doi.org/10.3390/ijerph18094668

Muratori, P., & Ciacchini, R. (2020). Children and the COVID-19 transition: Psychological reflections and suggestions on adapting to the emergency. *Clinical Neuropsychiatry, 17*(2), 131–134. https://doi.org/10.36131/CN20200219

Octavius, G. S., Silviani, F. R., Lesmandjaja, A., Angelina, & Juliansen, A. (2020). Impact of COVID-19 on adolescents' mental health: A systematic review. *Middle East Current Psychiatry, 27*(1). https://doi.org/10.1186/s43045-020-00075-4

O'Donovan, R., & Burke, J. (2022). Factors associated with post-traumatic growth in healthcare professionals: A systematic review of the literature. *Healthcare, 10*(12), 2524. https://doi.org/10.3390/healthcare10122524

Orille, A., Harrison, L., & Taku, K. (2019). Individual differences in attitudes and perceptions toward posttraumatic growth and illusory growth. *Personality and Individual Differences, 142*, 153–158. https://doi.org/10.1016/j.paid.2019.02.002

Panchal, U., Salazar de Pablo, G., Franco, M., Moreno, C., Parellada, M., Arango, C., & Fusar-Poli, P. (2021). The impact of COVID-19 lockdown on child and adolescent mental health: Systematic review. *European Child & Adolescent Psychiatry*. https://doi.org/10.1007/s00787-021-01856-w

Ramadhan, M. H. A., Putri, A. K., Melinda, D., Habibah, U., Fajriyah, U. N., Aini, S., … Ikhsan, D. S. (2020). Children's mental health in the time of COVID-19: How things stand and the aftermath. *Malaysian Journal of Medical Sciences, 27*(5), 196–201. https://doi.org/10.21315/mjms2020.27.5.15

Senejko, A., Godawa, G., Gurba, E., Kalus, A., & Gurba, K. (2022). Growth-related responses to the threat of COVID-19 among adolescents. *International Journal of Environmental Research and Public Health, 19*(19), Article 12597. https://doi.org/10.3390/ijerph191912597

Silvestre, M., & Tarquinio, C. L. (2022). Systemic family therapy and EMDR therapy: An integrative approach. *European Journal of Trauma & Dissociation, 6*(4), Article 100291. https://doi.org/10.1016/j.ejtd.2022.100291

Taddei, B., De Luca, L., & Nocentini, A. (2024). Posttraumatic growth trajectories among adolescents during COVID-19 pandemic: The role of time-varying stress reactions and systems' satisfaction. *Current Psychology, 43*(46), 35686–35696. https://doi.org/10.1007/s12144-024-06763-2

Tamiolaki, A., Kalaitzaki, A., Papadakaki, M., & Kourkoutas, E. (2024). The silver lining of posttraumatic growth around the dark side of the COVID-19 Pandemic: A school-based intervention with mindfulness and character strengths practices among children. *Healthcare, 12*(2), 283. https://doi.org/10.3390/healthcare12020283

Tang, W., Yan, Z., Lu, Y., & Xu, J. (2022). Prospective examination of adolescent emotional intelligence and post-traumatic growth during and after COVID-19 lockdown. *Journal of Affective Disorders, 309*, 368–374. https://doi.org/10.1016/j.jad.2022.04.129

Tedeschi, R. G., & Calhoun, L. G. (1996). The posttraumatic growth inventory: Measuring the positive legacy of trauma. *Journal of Traumatic Stress, 9*(3), 455–471. https://doi.org/10.1007/bf02103658

Tedeschi, R. G., & Calhoun, L. G. (2004). Target article: "posttraumatic growth: Conceptual foundations and empirical evidence. *Psychological Inquiry, 15*(1), 1–18. https://doi.org/10.1207/s15327965pli1501_01

Tedeschi, R. G., Shakespeare-Finch, J., Taku, K., & Calhoun, L. G. (2018). *Posttraumatic growth: Theory, research, and applications*. Routledge.

Ulset, V. S., & von Soest, T. (2022). Posttraumatic growth during the COVID-19 lockdown: A large-scale population-based study among Norwegian adolescents. *Journal of Traumatic Stress, 35*(3), 941–954. https://doi.org/10.1002/jts.22801

van der Hart, O., Steele, K., & Nijenhuis, E. (2017). The treatment of traumatic memories in patients with complex dissociative disorders. *European Journal of Trauma & Dissociation, 1*(1), 25–35. https://doi.org/10.1016/j.ejtd.2017.01.008

Waters, L., Allen, K.-A., & Arslan, G. (2020). Stress-related growth in adolescents returning to school after COVID-19 school closure. doi:10.31219/osf.io/ud7cz.

Wenter, A., Schickl, M., Sevecke, K., Juen, B., & Exenberger, S. (2022). Children's mental health during the first two years of the COVID-19 pandemic: Burden, risk factors and posttraumatic growth – A mixed-methods parents' Perspective. *Frontiers in Psychology, 13*. https://doi.org/10.3389/fpsyg.2022.901205

Xie, X., Xue, Q., Zhou, Y., Zhu, K., Liu, Q., Zhang, J., & Song, R. (2020). Mental health status among children in home confinement during the coronavirus disease 2019 outbreak in Hubei Province, China. *JAMA Pediatrics, 174*(9), 898–900. https://doi.org/10.1001/jamapediatrics.2020.1619

Xu, M., Tian, R., Fu, C., Li, J., Bi, D., & Wang, Y. (2023). The impact of family function on post-traumatic reactions of Chinese adolescents infected with COVID-19: A latent profile study. *Frontiers in Public Health, 11*. https://doi.org/10.3389/fpubh.2023.1153820

Yue, J., Zang, X., Le, Y., & An, Y. (2020). Anxiety, depression and PTSD among children and their parent during 2019 novel coronavirus disease (COVID-19) outbreak in China. *Current Psychology*. https://doi.org/10.1007/s12144-020-01191-4

Zeng, Z., Wang, X., Chen, Q., Gou, Y., & Yuan, X. (2022). Core belief challenge moderated the relationship between posttraumatic growth and adolescent academic burnout in Wenchuan area during the COVID-19 pandemic. *Frontiers in Psychology, 13*. https://doi.org/10.3389/fpsyg.2022.1005176

Zhang, A., Shi, L., Yan, W., Xiao, H., Bao, Y., Wang, Z., … Lu, L. (2021). Mental health in children in the context of COVID-19: Focus on discharged children. *Frontiers in Psychiatry, 12*. https://doi.org/10.3389/fpsyt.2021.759449

Zhen, B., Yao, B., & Zhou, X. (2022). How does parent–child communication affects posttraumatic stress disorder and growth in adolescents during the COVID-19 pandemic? The mediating roles of self-compassion and disclosure. *Journal of Affective Disorders, 306*, 1–8. https://doi.org/10.1016/j.jad.2022.03.029

Zhou, J., Wang, Y., Bu, T., Zhang, S., Chu, H., Li, J., … Yang, Y. (2021). Psychological impact of COVID-19 epidemic on adolescents: A large sample study in China. *Frontiers in Psychiatry, 12*. https://doi.org/10.3389/fpsyt.2021.769697

Zoellner, T., & Maercker, A. (2006). Posttraumatic growth and psychotherapy. In L. G. Calhoun, & R. G. Tedeschi (Eds.), *Handbook of posttraumatic growth: Research & practice* (pp. 334–354). Lawrence Erlbaum Associates Publishers.